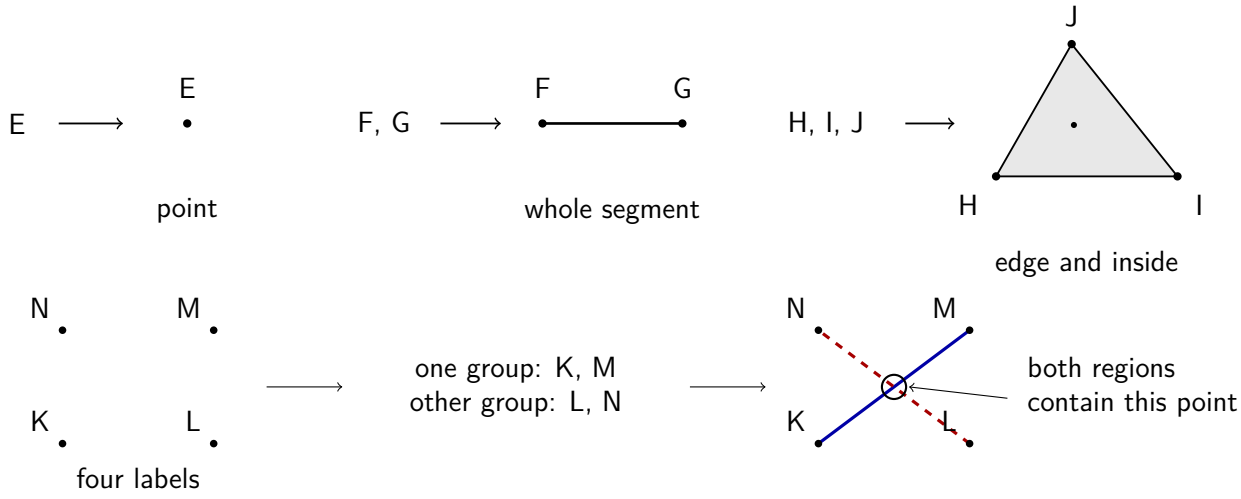
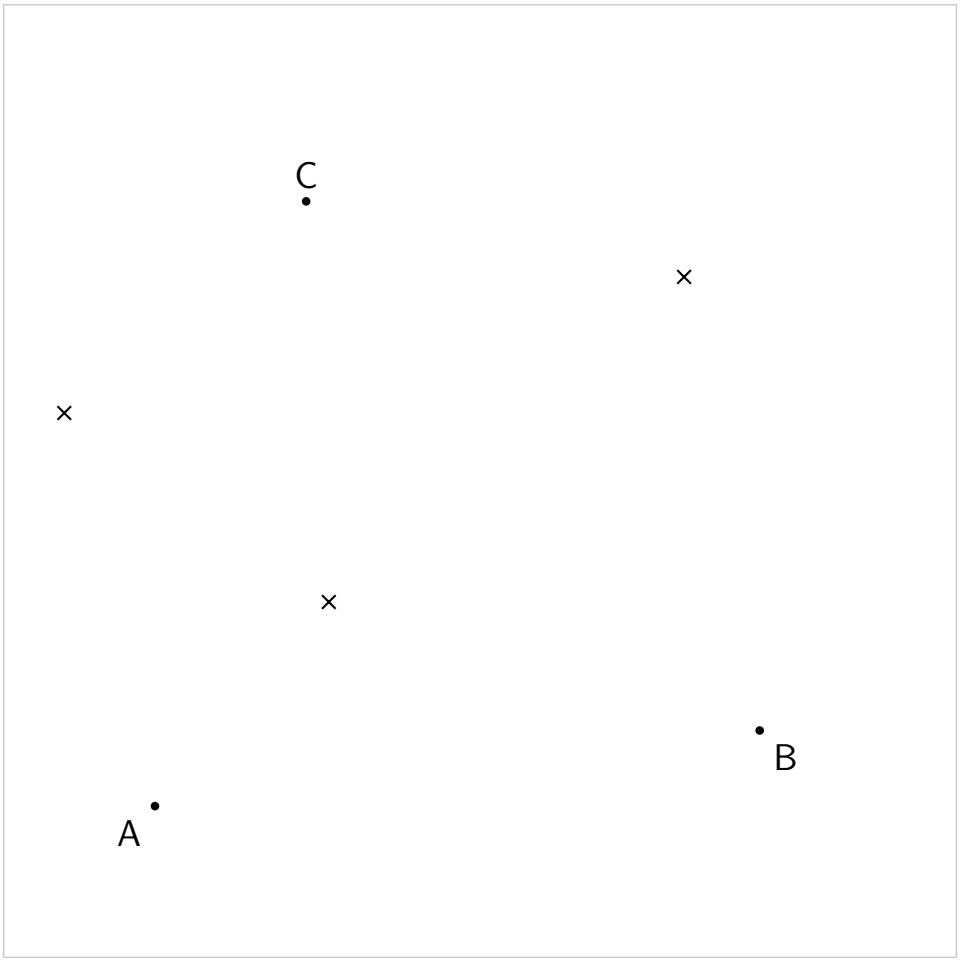
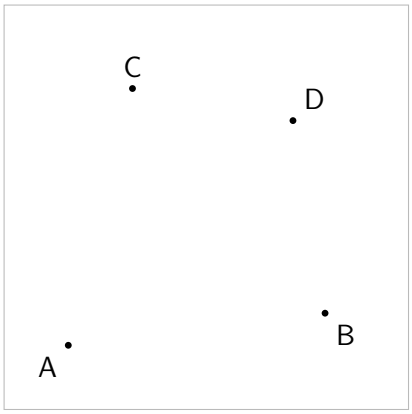
Use each label once in two groups, with at least one label in each. Draw each group's stretched-band region and lightly shade its inside. Use the centers of the marks. Sharing even one point counts. Swapping the two groups is the same split.

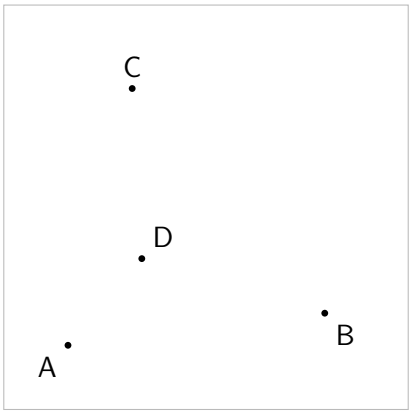


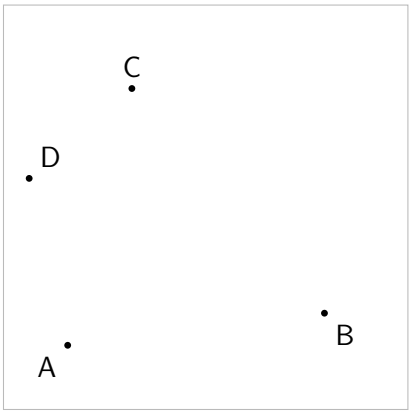
Problem 1: Put D on each cross. Each time, find a split whose regions meet.



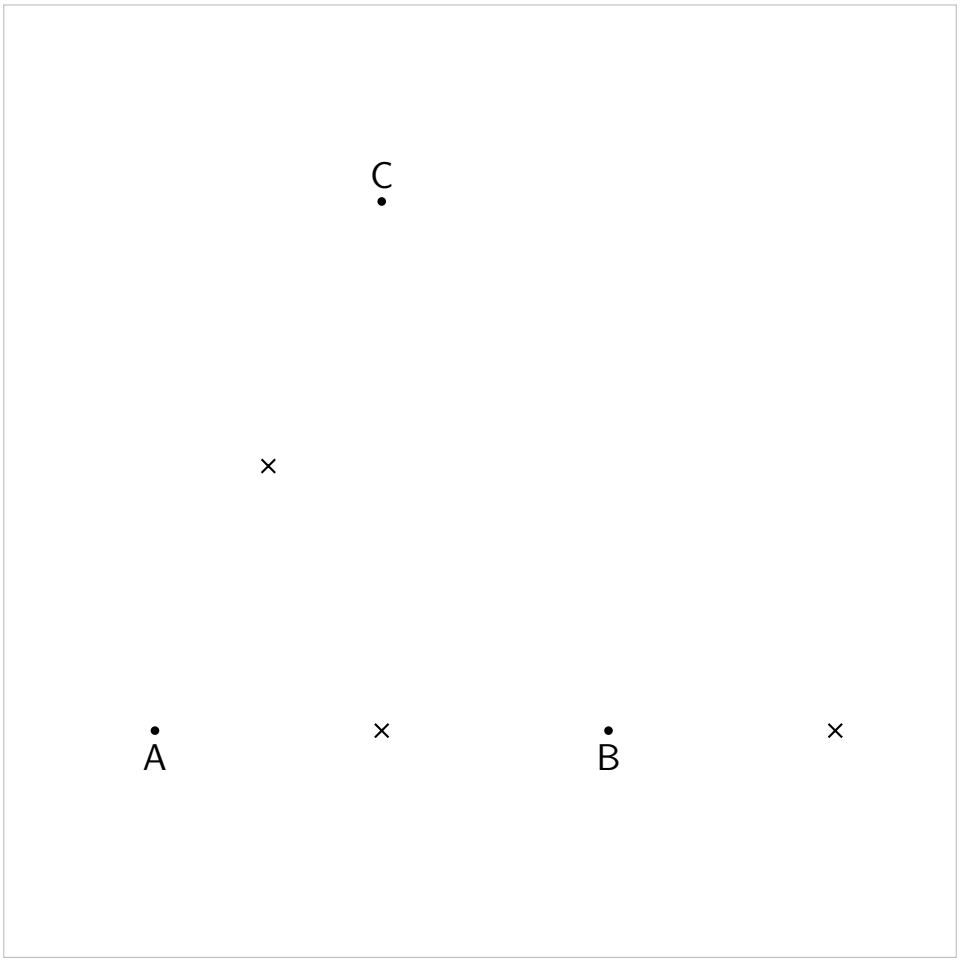
Problem 1: For each place of D, draw the two regions you found and mark a point they share.



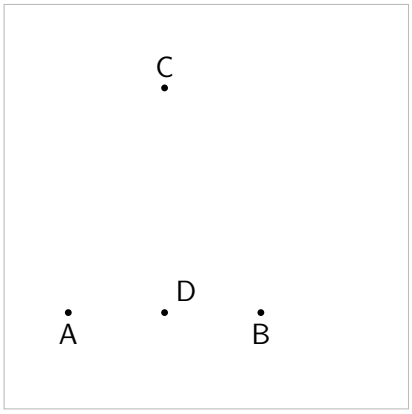


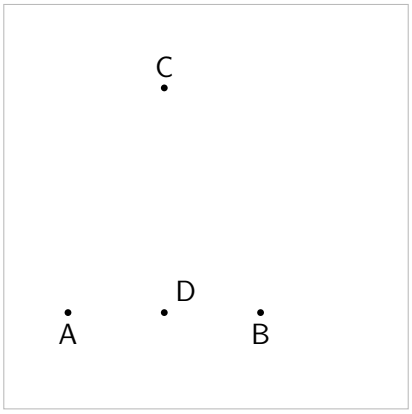


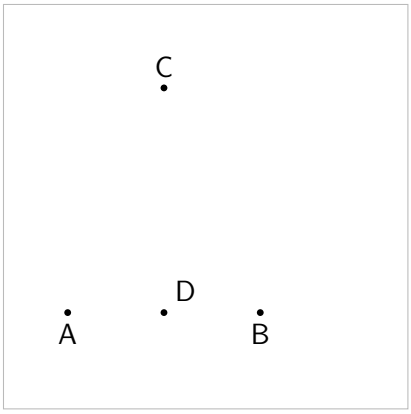
Problem 2: Try D on each cross. Find every split whose regions meet.

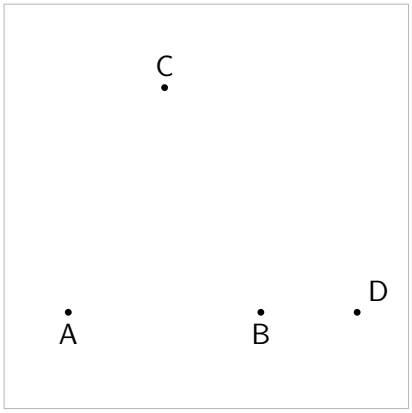


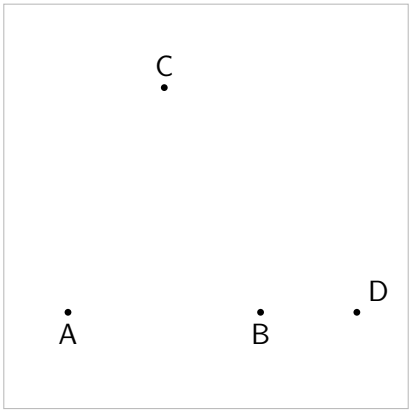
Problem 2: Find every split whose regions meet.

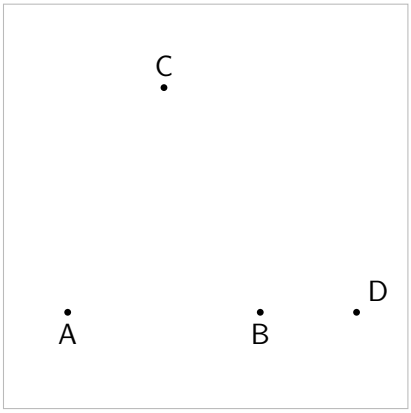


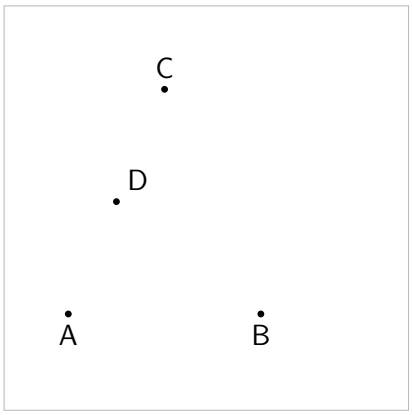


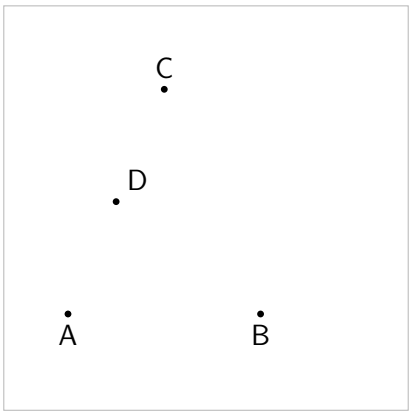


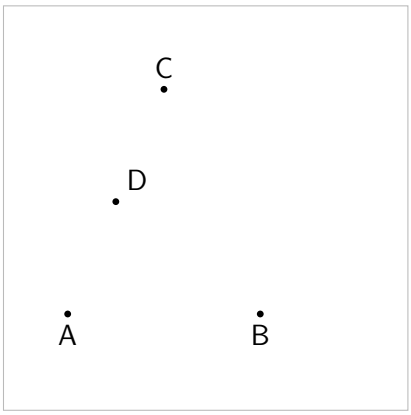




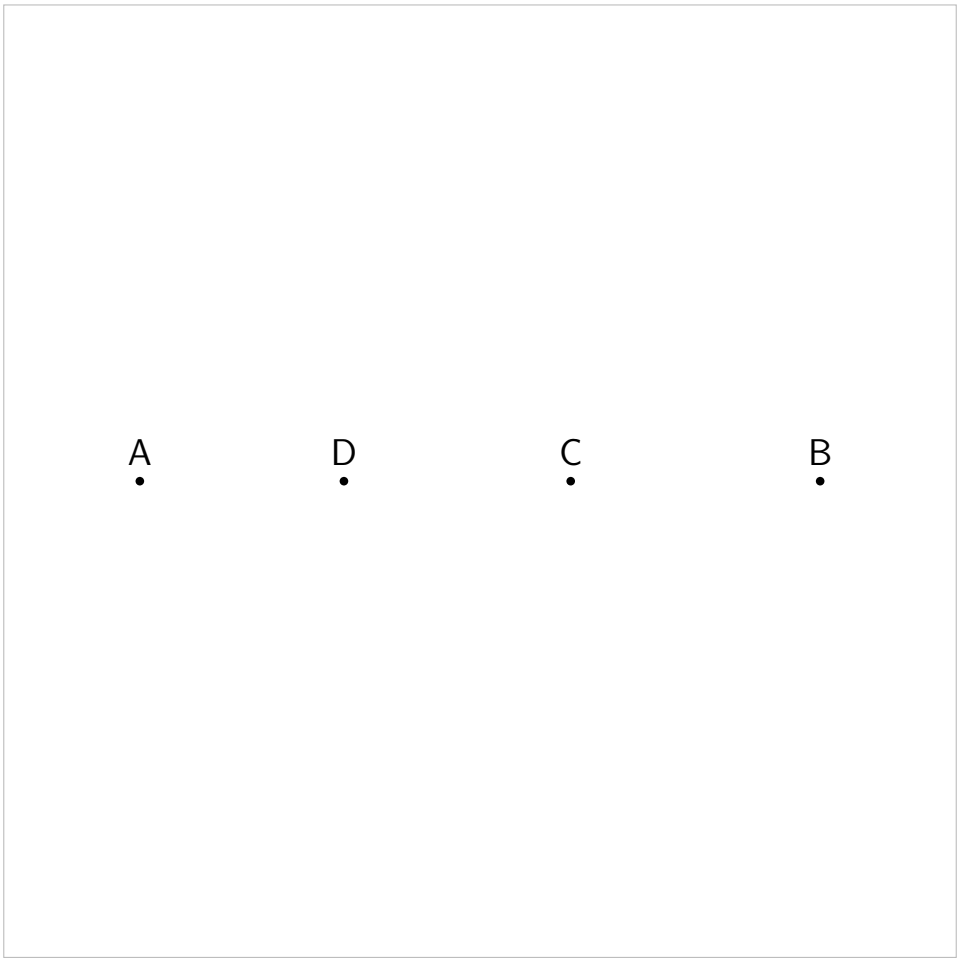




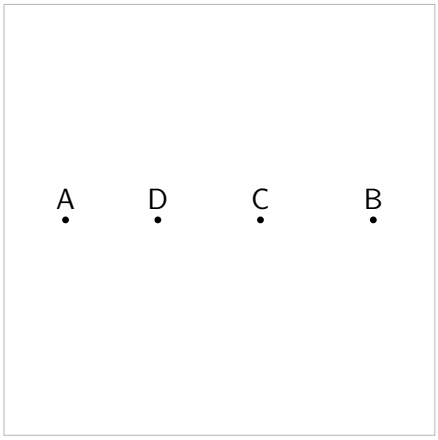


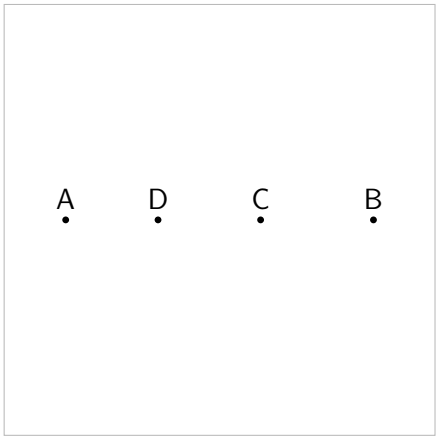


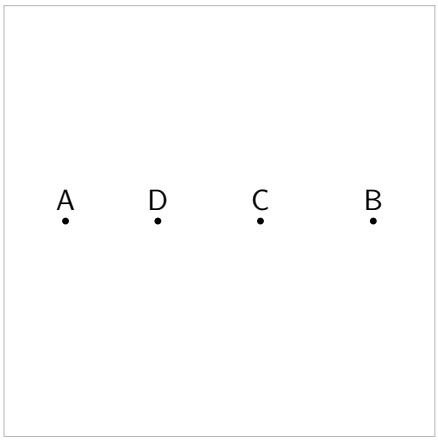
Problem 3: Find every split whose regions meet.

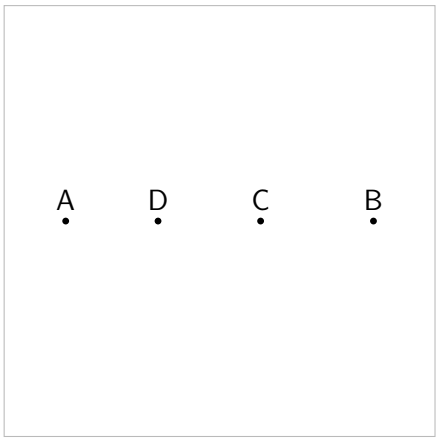


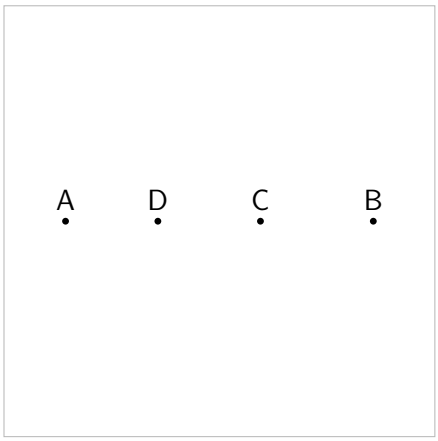
Problem 3: Find every split whose regions meet.

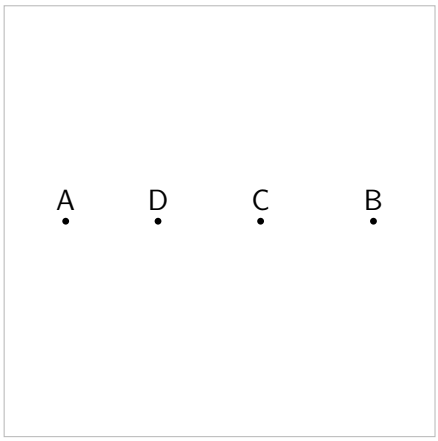




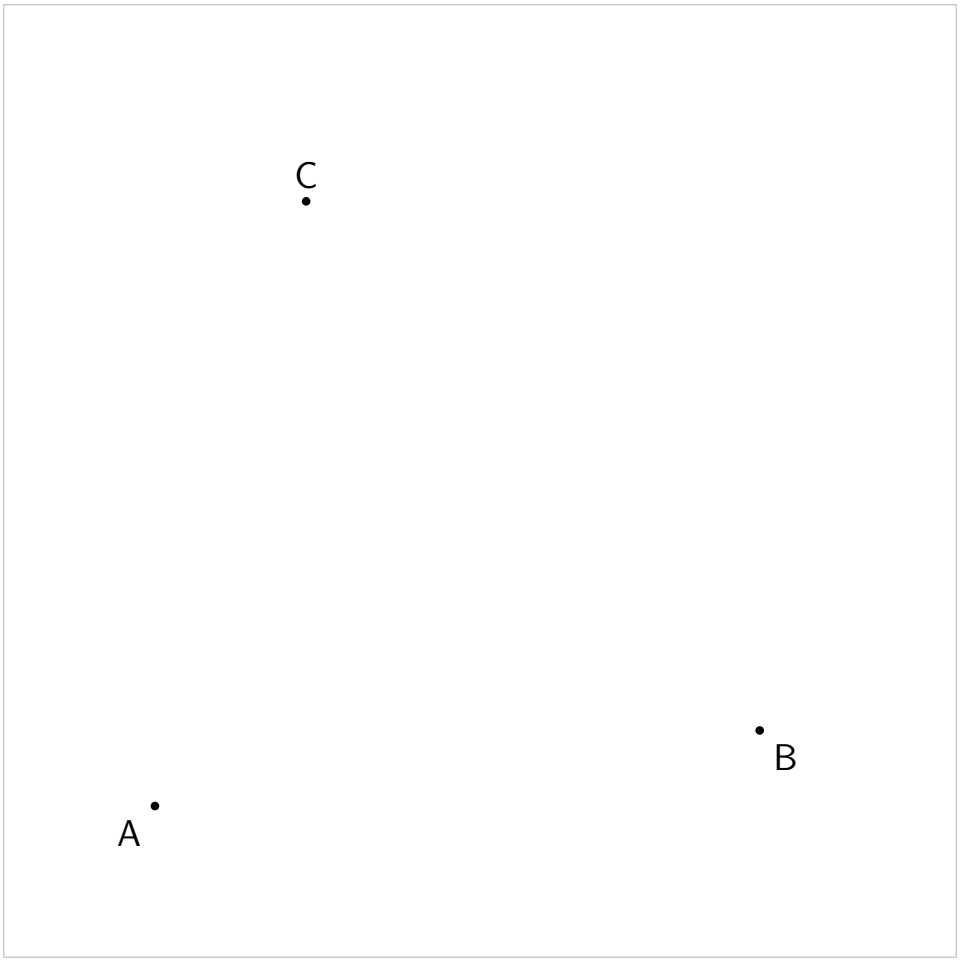




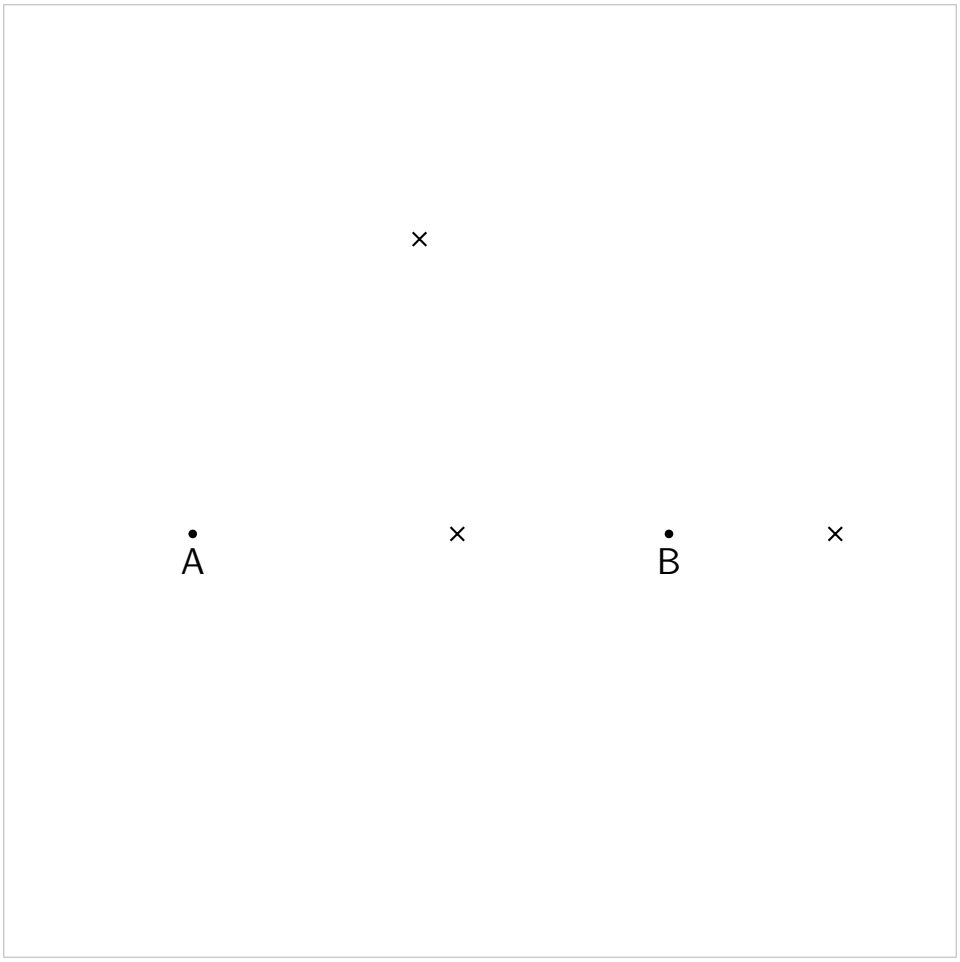




Problem 4: Keep A, B, and C fixed, and take turns placing D. Can the other player always make two regions that meet?



Problem 5: Keep A and B fixed, and try C on each cross. Find every place in the box where C can go so the two regions meet.



Problem 6: Take turns placing A, B, C, and D in the box. Can the other player always make two regions that meet?